NANJING UNIVERSITY

# Cloth Simulation

**Fang Hao**

**Nanjing University**

# Outline

- Motivation and Application
- Cloth Simulation Methods
  - Physically-based Cloth Simulation
- Reference

# Motivation & Application

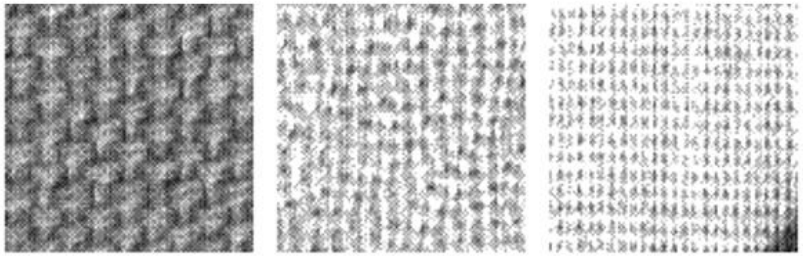
- Movies
- Games
- VR Scene
- Virtual Try-on
- Garment Design

# Cloth Simulation Methods

- Geometric Method
  - Represent folds and creases by geometrical equations
  - Aim at modeling the appearance of the cloth
  - Not focus on the physical aspects of cloth
- Physically-based Method

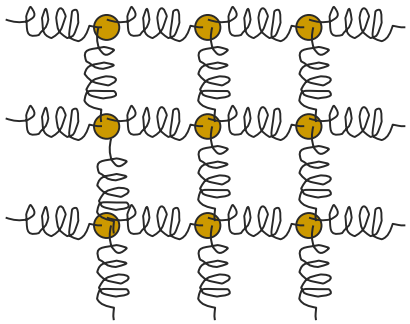
# Particle-based Model

- Cloth fibers and crossings

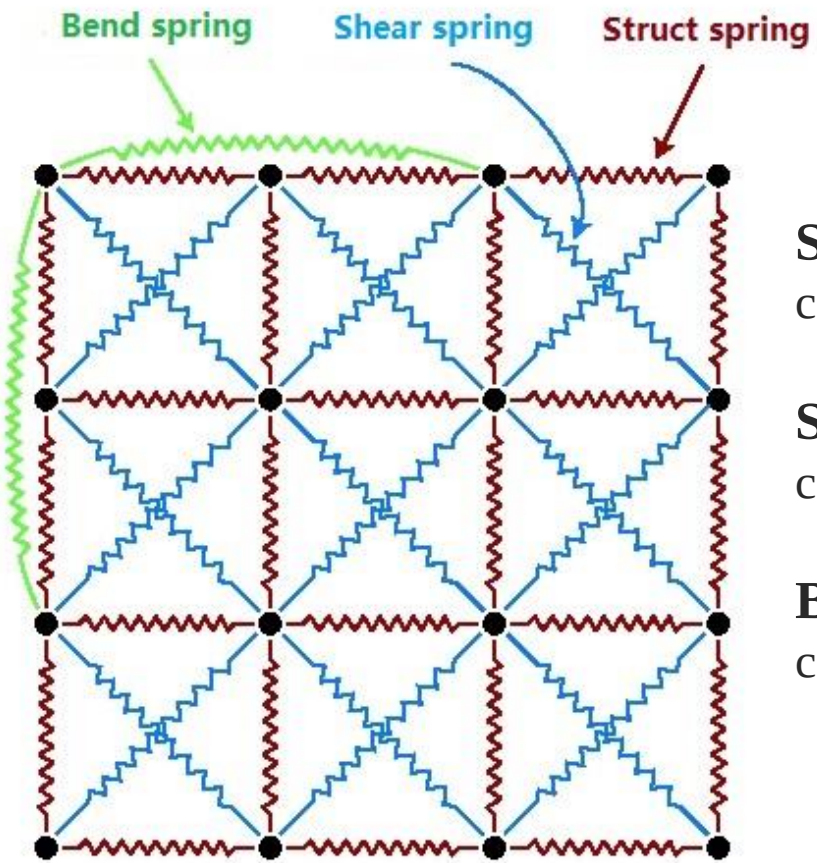


a) 100% cotton b) 100% wool c) polyester/cotton

- Approximated by mass-spring particle model



# Mass-spring Particle Model




**Struct spring:**
connect p[i,j] and p[i+1,j], p[i,j] and p[i,j+1]

**Shear spring:**
connect p[i,j] and p[i+1,j+1], p[i+1,j] and p[i,j+1]

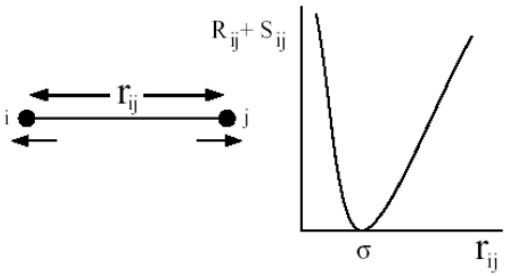
**Bend spring:**
connect p[i,j] and p[i+2,j], p[i,j] and p[i,j+2]

# Physically-based Simulation

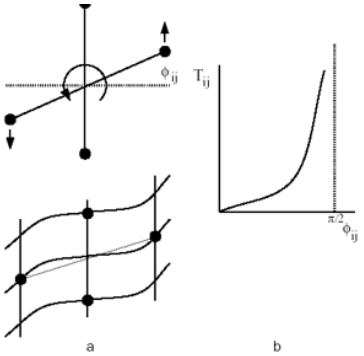
- 1. Finding Energy Equation
- 2. Solving the Equations
- 3. Collision Detection / Handling

# Energy Equation

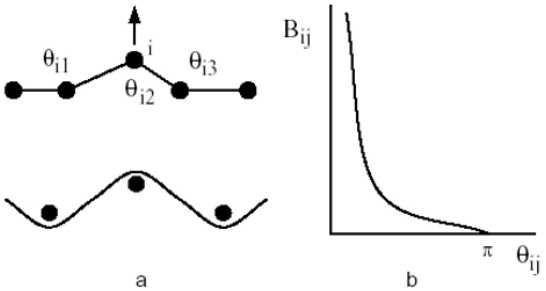
$$F = F_{internal} + F_{external} = F_{struct} + F_{shear} + F_{bend} + F_{wind} + F_{friction} + F_{gravity}$$




Struct




Shear




Bend

# Energy Equation

**Hooke's law:** $$Fstring(i,j) = -\sum k_{(i,j)}(|x_i - x_j|) - l_{(i,j)}\frac{x_i - x_j}{|x_i - x_j|}$$

$$Fwind = -\frac{1}{2}\rho |Vrel| cd\alpha (Vrel \bullet N) N$$

$$F_{fraction} = -K_{fraction} \bullet v_i$$

$$F_{fraction} = \mu g$$

# Solving the Equations

- Explicit Euler
- Implicit Euler
- Verlet integration
- Implicit integration

# Explicit Euler

$$v(t+\Delta t) = v(t) + a(t)\Delta t$$

$$x(t+\Delta t) = x(t) + v(t+\Delta t)$$

**X(t)/X(t+Δt)** : Point position at time **t / t+Δt**
**v(t)/v(t+Δt)** : Point speed at time **t / t+Δt**
**a(t)** : Point acceleration at time **t**

# Implicit Euler

$$v(t+\Delta t) = v(t) + a(t+\Delta \mathrm{t})\Delta t$$

$$x(t+\Delta t) = x(t) + v(t+\Delta t)$$

**X(t)/X(t+Δt)** : Point position at time **t / t+Δt**
**v(t)/v(t+Δt)** : Point speed at time **t / t+Δt**
**a(t+Δt)** : Point acceleration at time **t+Δt**

# Verlet integration

$$x(t+\Delta t) = x(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2 + \frac{1}{6}a(t)\Delta t^3 + o(\Delta t^4) \qquad (1)$$

$$x(t-\Delta t) = x(t) - v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2 - \frac{1}{6}a(t)\Delta t^3 + o(\Delta t^4) \qquad (2)$$

（1）+（2）：

$$x(t+\Delta t) = 2x(t) + x(t-\Delta t) + a(t)\Delta t^2 + o(\Delta t^4)$$

# Implicit integration

$$x(t+\Delta t) = 2x(t) - x(t-\Delta t) + a(t)\Delta t^2$$

$$v(t+\Delta t) = \frac{x(t+\Delta t) - x(t)}{\Delta t}$$

$$x(t+\Delta t) = x(t) + \frac{1}{2}(v(t) + v(t+\Delta t))\Delta t$$

# Collision Detection / Handling

- Collision Detection:
  - Cloth-cloth: particle-triangle and edge-edge intersection
  - Cloth-solid: cloth particle against the faces of solid object
- Collision Response:
  - Cloth-cloth: insert a strong damped spring force to push the cloth apart
  - Cloth-solid: if the relative tangential velocity is low,lock the particle onto the surface;if not allow the particle to slide on the surface

# References


- Meng Y，Mokb P Y，Jin X． Interactive virtual try-on clothing design systems［J］． Computer-Aided Design． 2010，42( 4) : 310-321
- Volino P， Magnenat-Thalmann N， Faiire F． A simple approach to nonlinear tensile stiffness for accurate cloth simulation［J］． ACM Transactions on Graphics，2009，28( 4) : 104．
- Zhang M M，Pan Z G，Mi Q R． Anisotropic cloth modeling for material fabric［J］． Transactions on Edutainment IV，2010( 4) : 130-144

# Online Demo

http://fanghao.site/OnlineDemos/Cloth-Simulation/index.html